PRISMA 2020 Screening Stage: Title and Abstract Screening Against Inclusion and Exclusion Criteria

Report prepared 8 August 2026

1. Scope and Review Topic

This screening pass was conducted for the systematic literature review titled "Concept Drift, Spatial Domain Shift and Continual Adaptation in Machine Learning-Based IoT Intrusion Detection: A Systematic Literature Review".

Eleven subsections of the review outline carry their own independent literature pool and were used as the screening targets against which the IC1 and EC1 criteria were judged, and against which the three-to-eight-record-per-subsection selection cap was applied, namely Section 1 (Introduction), Sections 3.1 to 3.3, Sections 4.1 to 4.3, Sections 5.1 to 5.3, and Section 6. Section 2.x (Review Methodology), Section 7 (Synthesis of Research Gaps), Sections 8.1 to 8.5 (Open Challenges), and Section 9 (Conclusion) are authored-protocol or synthesis-only chapters with no separate citation pool and were therefore excluded as screening targets.

Step 3 was run on the 3,089 unique records that survived Step 2 (deduplication).

2. Inclusion and Exclusion Criteria Applied

Table 1 and Table 2 report the inclusion (IC) and exclusion (EC) criteria applied in this screening run. These definitions were confirmed with the review team prior to screening and reflect the criteria in effect for this specific pass.

2.1 Inclusion Criteria

Table 1. Inclusion criteria applied in the Step 3 screening run

Code	Definition applied in this Step 3 run
IC1	The study proposes, evaluates, or surveys domains directly relevant to the SLR topic, referenced to a specific section or subsection.
IC2	The study is a foundational, domain-general contribution establishing a study captured under IC1, even where it predates the SLR topic or the IC5 date window.
IC3	The study is an original article or critical review article.
IC4	The study is published in English.
IC5	The study is published from 2020 to 2026 inclusive.
IC6	The full text is accessible online and the bibliographic details are verifiable.

2.2 Exclusion Criteria

Table 2. Exclusion criteria applied in the Step 3 screening run

Code	Definition applied in this Step 3 run
EC1	The study's primary focus lies outside the domains and themes reviewed in each section.
EC2	The study is a conference paper, book chapter, or conference report.
EC3	The study is not published in English.
EC4	The study is published outside 2020 to 2026, unless exempt as an IC2 foundational study.
EC5	The full text is inaccessible online or the bibliographic details are unverifiable.

3. Results

Of the 3,089 records screened against the criteria in Section 2, 141 records satisfied at least one inclusion criterion and no exclusion criterion and were therefore retained for full-text retrieval in Step 4. Table 3 summarizes the retained records by the inclusion criterion under which each was captured, and Table 4 lists each retained record individually.

Table 3. Records retained by inclusion criterion

Inclusion criterion	Records retained (n)
IC1	123
IC2	18
Total retained	**141**

The majority of retained records were captured under IC1, reflecting studies proposing, evaluating, or surveying domains directly relevant to the review topic. The remaining records were captured under IC2, comprising foundational, domain-general contributions, such as early work on dataset shift, concept drift detection, and continual learning, that established techniques or evaluation practices subsequently built upon by studies captured under IC1.

Table 4. Records retained following Step 3 screening

No.	IC	Title	Year
1	IC2	Internet of Things: A survey on machine learning-based intrusion detection approaches	2019
2	IC2	Network Intrusion Detection for IoT Security Based on Learning Techniques	2019
3	IC1	Recent endeavors in machine learning-powered intrusion detection systems for the Internet of Things	2024
4	IC1	A comprehensive survey on deep learning-based intrusion detection systems in Internet of Things (IoT)	2025
5	IC1	A Survey of Deep Learning Technologies for Intrusion Detection in Internet of Things	2024
6	IC1	A Taxonomy of Machine-Learning-Based Intrusion Detection Systems for the Internet of Things: A Survey	2022
7	IC2	A survey on transfer learning	2010
8	IC2	Improving predictive inference under covariate shift by weighting the log-likelihood function	2000

No.	IC	Title	Year
9	IC2	A unifying view on dataset shift in classification	2012
10	IC1	Preventing dataset shift from breaking machine-learning biomarkers	2021
11	IC1	A Survey of Unsupervised Deep Domain Adaptation	2020
12	IC1	A Comprehensive Survey on Test-Time Adaptation Under Distribution Shifts	2025
13	IC1	EFFICIENT AND MULTIPLY ROBUST RISK ESTIMATION UNDER GENERAL FORMS OF DATASET SHIFT	2024
14	IC1	When Invariant Representation Learning Meets Label Shift: Insufficiency and Theoretical Insights	2024
15	IC1	Evaluation of domain generalization and adaptation on improving model robustness to temporal dataset shift in clinical medicine	2022
16	IC1	A unified framework for dataset shift diagnostics	2023
17	IC2	Learning under Concept Drift: A Review	2019
18	IC2	A survey on concept drift adaptation	2014
19	IC2	Characterizing concept drift	2016
20	IC1	One or two things we know about concept drift—a survey on monitoring in evolving environments. Part A: detecting concept drift	2024
21	IC1	From concept drift to model degradation: An overview on performance-aware drift detectors	2022
22	IC1	Stream-learn — open-source Python library for difficult data stream batch analysis	2022
23	IC1	A survey of processing methods for different types of concept drift	2026
24	IC1	Concept Drift Analysis Based on Isolation Forest for Effectively Detecting Network Attack in IoT Scenarios	2026
25	IC2	Table for estimating the goodness of fit of empirical distributions	1948
26	IC2	Learning with drift detection	2004
27	IC2	Multiple Comparisons among Means	1961
28	IC1	Detection of data drift in a two-dimensional stream using the Kolmogorov-Smirnov test	2022
29	IC1	Reactive Soft Prototype Computing for Concept Drift Streams	2020
30	IC1	Intrusion detection in the IoT data streams using concept drift localization	2024
31	IC1	Diversity measure as a new drift detection method in data streaming	2020
32	IC1	A Representation Learning Approach to Feature Drift Detection in Wireless Networks	2026
33	IC1	A Network Intrusion Detection System for Concept Drifting Network Traffic Data	2021
34	IC1	XDFC-IDS: An explainable decentralized federated class-incremental fusion framework for intrusion detection	2026
35	IC1	SFC-NIDS: a sustainable and explainable flow filtering based concept drift-driven security approach for network introspection	2024
36	IC1	Class imbalance and concept drift invariant online botnet threat detection framework for heterogeneous IoT edge	2024
37	IC1	Self-Supervised Adaptation Method to Concept Drift for Network Intrusion Detection	2025
38	IC1	Intrusion Detection in the IoT Under Data and Concept Drifts: Online Deep Learning Approach	2022

No.	IC	Title	Year
39	IC1	Unsupervised Incremental Learning with Dual Concept Drift Detection for Identifying Anomalous Sequences	2024
40	IC1	Addressing Concept Drift in IoT Anomaly Detection: Drift Detection, Interpretation, and Adaptation	2024
41	IC1	Feature Drift Aware for Intrusion Detection System Using Developed Variable Length Particle Swarm Optimization in Data Stream	2023
42	IC1	From one attack domain to another: Contrastive transfer learning with siamese networks for APT detection	2025
43	IC1	Reinforcement Learning-Based Voting for Feature Drift-Aware Intrusion Detection: An Incremental Learning Framework	2025
44	IC1	Synthetic Non-stationary Data Streams for Recognition of the Unknown	2026
45	IC1	Concept Drift Detection in Data Stream Mining: A literature review	2022
46	IC1	Challenges in benchmarking stream learning algorithms with real-world data	2020
47	IC2	Evolving fuzzy-rule-based classifiers from data streams	2008
48	IC2	The impact of diversity on online ensemble learning in the presence of concept drift	2010
49	IC2	Incremental on-line learning: A review and comparison of state of the art algorithms	2018
50	IC1	Enhanced Intrusion Detection with Data Stream Classification and Concept Drift Guided by the Incremental Learning Genetic Programming Combiner	2023
51	IC1	Concept Drift–Based Intrusion Detection For Evolving Data Stream Classification In IDS: Approaches And Comparative Study	2024
52	IC1	IL-IDS: an incremental learning approach with confined data streams for intrusion detection	2025
53	IC1	An Incremental Learning Method Based on Dynamic Ensemble RVM for Intrusion Detection	2022
54	IC2	Learning without Forgetting	2018
55	IC2	Overcoming catastrophic forgetting in neural networks	2017
56	IC2	Continual lifelong learning with neural networks: A review	2019
57	IC1	Three types of incremental learning	2022
58	IC1	Continual Learning With Knowledge Distillation: A Survey	2025
59	IC1	A Continual Learning Survey: Defying Forgetting in Classification Tasks	2022
60	IC1	Addressing catastrophic forgetting in class-incremental learning—a survey	2026
61	IC1	Overcoming Class Imbalance in Incremental Learning Using an Elastic Weight Consolidation-Assisted Common Encoder Approach	2025
62	IC1	Non-IID data and Continual Learning processes in Federated Learning: A long road ahead	2022
63	IC1	Federated continual learning: A comprehensive survey on lifelong and privacy-preserving learning over distributed and non-stationary data	2026
64	IC1	Advances and open problems in federated learning	2021
65	IC1	FD-IDS: Federated Learning with Knowledge Distillation for Intrusion Detection in Non-IID IoT Environments	2025
66	IC1	Dataset-centric evaluation of federated intrusion detection models in IoT networks	2026
67	IC1	FORT-IDS: a federated, optimized, robust and trustworthy intrusion detection system for IIoT security	2026

No.	IC	Title	Year
68	IC1	FL-MalDrift: a federated learning framework for malware detection under local concept drift	2026
69	IC1	Evaluating the Generalization Gaps of Intrusion Detection Systems Across DoS Attack Variants	2025
70	IC1	Benchmarking and Cross-Dataset Evaluation of AI-Based Intrusion Detection Systems for Smart City IoT Networks	2026
71	IC1	Inter-dataset generalization strength of supervised machine learning methods for intrusion detection	2020
72	IC1	DI-NIDS: Domain invariant network intrusion detection system	2023
73	IC1	Explainable Cross-domain Evaluation of ML-based Network Intrusion Detection Systems	2023
74	IC1	A Survey of Machine and Deep Learning Methods for Internet of Things (IoT) Security	2020
75	IC1	Machine Learning in IoT Security: Current Solutions and Future Challenges	2020
76	IC1	Concept Drift Under Harsh Constraints: A Review of Potential Strategies for IoT Systems	2025
77	IC1	A Comprehensive Survey on Concept-Drift-Resilient Network Intrusion Detection Systems	2026
78	IC1	Handling Out-of-Distribution Data: A Survey	2025
79	IC1	Online kernel CUSUM for change-point detection	2026
80	IC1	CDDA-MD: An efficient malicious traffic detection method based on concept drift detection and adaptation technique	2025
81	IC1	Drift Adaptive Online DDoS Attack Detection Framework for IoT System	2024
82	IC1	Concept Drift Analysis by Dynamic Residual Projection for Effectively Detecting Botnet Cyber-Attacks in IoT Scenarios	2022
83	IC1	A Survey of Data Stream-Based Intrusion Detection Systems	2025
84	IC1	A comparative study on online machine learning techniques for network traffic streams analysis	2022
85	IC1	ATTA-FL-Lite: Lightweight Byzantine-Robust Federated Learning for Resource-Constrained Medical IoT Devices	2026
86	IC1	Secure and Explainable Federated Learning for IoT Intrusion Detection: A Comprehensive Survey	2026
87	IC2	MOA: Massive online analysis	2010
88	IC1	Machine Learning-Enabled IoT Security: Open Issues and Challenges under Advanced Persistent Threats	2023
89	IC1	Deep learning for intrusion detection in emerging technologies: a comprehensive survey and new perspectives	2025
90	IC1	Network intrusion datasets: A survey, limitations, and recommendations	2025
91	IC1	A Review on Machine Learning and Deep Learning Perspectives of IDS for IoT: Recent Updates, Security Issues, and Challenges	2021
92	IC1	A critical review of intrusion detection systems in the internet of things: techniques, deployment strategy, validation strategy, attacks, public datasets and challenges	2021
93	IC1	Adversarial Machine Learning for Network Intrusion Detection Systems: A Comprehensive Survey	2023
94	IC1	Machine learning based solutions for security of Internet of Things (IoT): A survey	2020

No.	IC	Title	Year
95	IC1	Generalization Error Bounds for Graph Neural Networks in Unseen Domains	2025
96	IC1	Towards Accurate and Robust Domain Adaptation Under Multiple Noisy Environments	2023
97	IC1	On the value of label and semantic information in domain generalization	2023
98	IC1	Dynamical Targeted Ensemble Learning for Streaming Data With Concept Drift	2024
99	IC1	Concept Drift Adaptation by Exploiting Drift Type	2024
100	IC1	Type-LDD: A Type-Driven Lite Concept Drift Detector for Data Streams	2024
101	IC1	Adaptation Strategies for Automated Machine Learning on Evolving Data	2021
102	IC1	Ensemble Approaches for Dynamic Data Stream Classification Under Label Scarcity	2025
103	IC1	Bayesian Nonparametric Unsupervised Concept Drift Detection for Data Stream Mining	2020
104	IC1	Novel hybrid pair recommendations based on a large-scale comparative study of concept drift detection	2021
105	IC1	N-DD: New Approach for Drift Detection Based on Neutrosophic Support Vector Machine	2025
106	IC1	An Integrated Preprocessing and Drift Detection Approach With Adaptive Windowing for Fraud Detection in Payment Systems	2025
107	IC1	Adaptive telecom churn prediction for concept-sensitive imbalance data streams	2022
108	IC1	Anomalies detection using isolation in concept-drifting data streams	2021
109	IC1	Real-Time Intrusion Detection Using Adaptive Sliding Windows for Concept Drift	2025
110	IC1	ADDAEIL: Anomaly Detection with Drift-Aware Ensemble-Based Incremental Learning	2025
111	IC1	Fast & Furious: On the modelling of malware detection as an evolving data stream[Formula presented]	2023
112	IC1	DFAID: Density-aware and feature-deviated active intrusion detection over network traffic streams	2022
113	IC1	FADES: Adaptive Drift Estimation via Conformal Signals for Streaming Intrusion Detection	2026
114	IC1	VLSA-CL: A variational latent space alignment and contrastive learning framework for robust detection of IoT botnet attacks under concept drift	2026
115	IC1	A NOVEL METHOD FOR DRIFT DETECTION IN STREAMING DATA BASED ON MEASUREMENT OF CHANGES IN FEATURE RANKS	2024
116	IC1	A Framework for Drift Detection and Adaptation in AI-driven Anomaly and Threat Detection Systems	2025
117	IC1	DyP-CNX: A Dynamic Preprocessing-Enhanced Hybrid Model for Network Intrusion Detection	2025
118	IC1	Evolving cybersecurity frontiers: A comprehensive survey on concept drift and feature dynamics aware machine and deep learning in intrusion detection systems	2024
119	IC1	Concept drift detection in district heating: A characterization framework and detector-centric benchmark	2026
120	IC1	Concept anchor drift detection: a framework for proactive detection of abrupt drifts in real-world data streams	2026
121	IC1	Concept drift detection methods based on different weighting strategies	2024

No.	IC	Title	Year
122	IC1	DRIFTNET-EnVACK: Adaptive Drift Detection in Cloud Data Streams With Ensemble Variational Auto-Encoder Featuring Contextual Network	2024
123	IC1	A Four-stage Bayesian Incremental Learning Framework for Intrusion Detection in Real-world IoT Environment	2026
124	IC1	Active Weighted Aging Ensemble for drifted data stream classification	2023
125	IC1	An online ensemble classification algorithm for multi-class imbalanced data stream	2024
126	IC1	Unsupervised feature selection for outlier detection on streaming data to enhance network security	2021
127	IC1	Class-incremental learning of convolutional neural networks based on double consolidation mechanism	2020
128	IC1	Class Incremental Learning with probability dampening and cascaded gated classifier	2025
129	IC1	Variance-Covariance Regularization Improves Continual Learning	2025
130	IC1	Towards anti-forgetting with masked optimal transport regularization for continual named entity recognition	2026
131	IC1	Communication-Efficient Federated Class-Incremental Intrusion Detection for Edge IoT Networks	2026
132	IC1	Non-IID and aware federated intrusion detection with PBFT with secured model aggregation for multi institutional healthcare internet of things networks	2026
133	IC1	Adaptive Hierarchical Federated Learning for IoT Anomaly Detection	2025
134	IC1	Federated continual learning with domain-routed historical expert matching for multimodal perception	2026
135	IC1	Fed-CALiBER: Federated Lightweight BERT Intrusion Detection on CAN Bus Protocol in Autonomous Vehicle	2025
136	IC1	Federated learning in intrusion detection: advancements, applications, and future directions	2025
137	IC1	A Systematic Review of Intrusion Detection Systems for Internet of Medical Things: Performance, Efficiency, Explainability, and Generalization	2026
138	IC1	Toward Generalization and Interpretable Deep Learning-Based Intrusion Detection System for Heterogeneous Network Environments	2026
139	IC1	Machine Learning in Network Intrusion Detection: A Cross-Dataset Generalization Study	2024
140	IC1	SHAP-Guided Feature Selection for Cross-Dataset Generalization in Network Intrusion Detection Systems	2026
141	IC1	A Decentralized Federated WGAN-GP Framework for Anomaly Detection in Industrial Control Systems	2026

This retained set of 141 records constitutes the input to Step 4, in which the online accessibility of each full text is verified prior to full-text retrieval.

4. Data Integrity Notes

Each record was screened on its title and abstract only, against the inclusion and exclusion criteria defined in Section 2. A record was retained where it satisfied at least one inclusion criterion and no exclusion criterion. Consistent with the PRISMA 2020 protocol, no full-text

assessment was undertaken at this step, that assessment forms Step 4 of the review pipeline following full-text retrieval.

5. Document Verification

This report has been prepared, checked, and approved in accordance with the review team's internal quality assurance procedure prior to proceeding to Step 4.

Prepared by	Checked by	Approved by
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